

Use of Grid-Forming Inverters in the Power Grid

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Abstract – This paper is aimed to present information on grid-forming (GFM) inverters for power system applications. With the popularity of renewable energy (RE) systems, power electronics have grown more prominent within power systems. When RE systems are connected to the main grid, inverters are required. In the following, grid-connected inverter operation and capabilities are discussed. Typical use of GFM inverters within power systems is also addressed.

I. INTRODUCTION

Conventional generation, such as steam and gas turbine power plants, makes up a large portion of global energy generation. However, due to the increase in public awareness for the environment, there has been a global shift toward renewable energy (RE) generation [1]-[2]. A significant amount of RE sources produce direct current (DC), while alternating current (AC) is utilized on the main grid. Due to the increase in RE generation, robust and dependable inverter technology is required to convert DC into AC [3]. Power grid inverters are commonly utilized in grid-following (GFL) or grid-forming (GFM) modes [4]. The GFL inverter acts like a current source, while a GFM inverter acts like a voltage source [2], [5]. Conventional inverters are generally GFL inverters, used to maximize power production from RE generation and to send quality power to main grids [5]-[6]. With these conventional GFL inverters comes issues with stability and vulnerabilities to grid variations [2]. With GFM inverters, there is better control over unbalanced voltages and improved inertia characteristics [1]-[2],

[7]. The rest of this report will investigate GFM inverters because of these advantages.

II. BASICS

This section discusses the background information of grid inverters and some GFM functions. The basic idea of an inverter is to turn DC into AC. It is important that grid-connected inverters are designed with extra precautions, so they follow power grid regulations. In Figure 1, a typical block diagram of a renewable energy generation system is shown. The renewable energy sources are connected to a grid-connected inverter, combining into a unit connection (PGUC) [8]. These PGUCs are then connected to a substation transformer, combining into a point of connection (POC) [8]. These generation stations are then connected with transmission lines to a point of common coupling (PCC) [8]. Within the renewable energy generation system, this report is most concerned with the grid-connected inverters, also outlined in red in Figure 1.

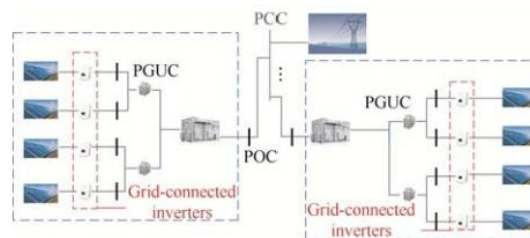


Figure 1. Renewable energy generation system [8]

As described earlier, the GFM inverter behaves like a voltage source. In Figure 2, a block diagram of a grid-forming inverter in grid-connected operation [2]. $V^*(f^*)$ is the output voltage of the GFM inverter

and Z_s is the equivalent impedance of the GFM inverter [2].

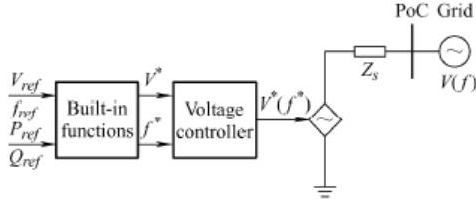


Figure 2. Grid-forming inverter connected to grid [2]

The GFM inverter output controls voltage magnitude, frequency, and phase angle, which can influence the active and reaction power of the grid [2]. To obtain this desired output of the GFM inverter, P_{ref} and Q_{ref} are sent to the built-in functions, leading to modified references V^* and f^* for the voltage controller [2].

Figure 3 demonstrates a GFL inverter that is connected to the grid. The GFL inverter is modeled with a current source instead of a voltage source that a GFM inverter is modeled with [2].

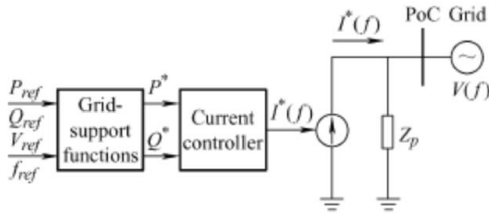


Figure 3. Grid-following inverter connected to grid [2]

This configuration modifies power references P^* and Q^* and generates a $I^*(f)$ [2]. In GFL operation, current magnitude and phase angle of the GFL output are adjusted to obtain the desired active/reactive power outputs, with frequency controlled by the grid [2].

III. CAPABILITIES

GFM inverters have better control over unbalanced voltages, better inertia characteristics, and frequency control.

A. Unbalanced Voltages

In power systems, voltage unbalances can come from unsymmetrical loads and power system faults. GFL inverters rely on an exterior voltage source, and have limited ability when met with unbalanced voltages [2], [7]. GFM inverters can control their voltages and phases, which gives them the ability to reduce negative sequence voltage [7]. This makes GFM inverters superior over GFL inverters when facing unbalanced voltages or loads.

B. Inertia

Inertia in a power system is how much resistance a power grid is affected by sudden changes of grid frequency (ie. a generator stopping). GFL inverters have zero-inertia characteristics, which lowers the overall inertia of the power system [2]. With a GFM inverter, they have the function to behave like a traditional synchronous machine (SM), providing synthetic inertia to the grid [1]. GFM inverters are helpful for power systems in terms of inertia, overall helping with grid stability.

C. Frequency

Frequency is important for any power system. There are usually tight regulations for operating frequency that a grid must operate at. This is to protect systems that are connected to the grid from damage. GFM inverters, unlike GFL inverters, can control the frequency output [2]. This is important because if a grid is operating at a non-nominal frequency, GFM can correct the grids frequency.

IV. APPLICATIONS

A. Microgrid

With the increase in distributed energy resources, microgrids (MGs) have grown more popular in order to improve power transfer efficiency [10]. The GFM inverter may be enabled in an islanded operation, or in other words, support a regional electric grid [2]. This allows the grid to continue to operate in the case of grid emergencies, adding to GFM stability traits [10].

B. Grid-Connected Inverter Networks

It is expected that with a large amount of RE sources, multiple inverters will be used to convert DC to AC. In [4], they propose a single GFM inverter to be used in an RE system. This is to maintain synchronization for GFL inverters and to ensure proper inertia [4]. A network with only GFM inverters is not recommended for its challenges related to synchronization, circulating reactive powers, unregulated power sharing, etc. [11]. Still, a GFM inverter is needed as the use of only GFL inverters will have no grid to follow and to synchronize to [2] [4].

C. Black Start

Blackouts are a crucial consideration of power systems. The black-start function allows grid to restore power without any support of external grids [2]. This can be done with single or multiple GFM inverters, as long as it supports the initial voltage, frequency, inertia, and etc. [2]. Since GFL inverters depend on existing strong grids, they do not have the black-start function [1]. However, after the buildup of the initial voltage from the GFM inverters, other DER inverters and generators can be connected to resume normal operation of the microgrid [2]. These systems are still future work and require more investigation to ensure real applications [9].

In Figure 4, an illustration of a GFM inverter is shown while in independent operation [2]. Under black-start function, this is more what the system looks like. In this case, Z_{load} may be the load of the microgrid, or the part of the grid the black-start is attempting to give power to.

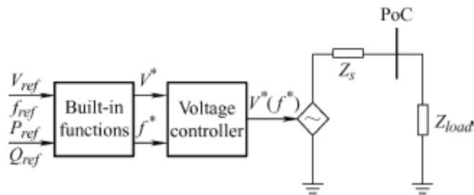


Figure 4. Grid-forming inverter in independent operation [2]

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